

Homotopy Invariance and First Computations

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1 Singular Chains and Prism Triangulations

Singular chains are free abelian on continuous simplices. A triangulation of a simplex times an interval converts a homotopy into a homomorphism of degree one.

1.1 Singular Chains and Chain Homotopies

We use integer coefficients and put singular chain groups in negative degrees equal to zero. Write Top for topological spaces and continuous maps, and Ab for abelian groups. For $n \geq 0$, the standard simplex is the following subspace of $\mathbb{R}^{n+1}$:

$$|\Delta^n| = \left\{ (t_0, \dots, t_n) : t_i \geq 0, \sum_{i=0}^n t_i = 1 \right\}.$$

Its vertices are $e_0, \dots, e_n$, and a point has the unique expression $x = \sum_{i=0}^n t_i e_i$. The numbers t_i are its barycentric coordinates. A singular n -simplex of X is a continuous map

$$\sigma : |\Delta^n| \rightarrow X.$$

The group $C_n(X)$ is free abelian on these maps. We also write σ for its generator in $C_n(X)$.

For $n \geq 1$, let i denote a base face index with $0 \leq i \leq n$. Define the affine face inclusion by

$$d^i : |\Delta^{n-1}| \rightarrow |\Delta^n|, \quad (t_0, \dots, t_{n-1}) \mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}).$$

The differential is the homomorphism specified on generators by

$$\partial_n : C_n(X) \rightarrow C_{n-1}(X), \quad \sigma \mapsto \sum_{i=0}^n (-1)^i \sigma d^i \quad (n \geq 1).$$

The degree-zero differential is the zero homomorphism

$$\partial_0 : C_0(X) \rightarrow 0.$$

The face identities give $\partial_{n-1}\partial_n = 0$. Define

$$H_n(X) = \ker(\partial_n) / \text{im}(\partial_{n+1}).$$

A continuous map and its induced maps are

$$f : X \rightarrow Y, \quad f_{\#,n} : C_n(X) \rightarrow C_n(Y), \quad \sigma \mapsto f\sigma,$$

$$f_{*,n} : H_n(X) \rightarrow H_n(Y), \quad [z] \mapsto [f_{\#,n}(z)].$$

Postcomposition commutes with faces, so $f_{\#}$ is a chain map. We omit degree subscripts when the source and target specify them.

Definition 1.1. Let $B_{\bullet}$ and $C_{\bullet}$ be chain complexes with differentials

$$d_{B,n} : B_n \rightarrow B_{n-1}, \quad d_{C,n} : C_n \rightarrow C_{n-1}.$$

A chain homotopy between chain maps

$$f, g : B_{\bullet} \rightarrow C_{\bullet}$$

is a family of homomorphisms

$$K_n : B_n \rightarrow C_{n+1} \quad (n \in \mathbb{Z})$$

satisfying, in each degree,

$$g_n - f_n = d_{C,n+1}K_n + K_{n-1}d_{B,n} : B_n \rightarrow C_n.$$

Proposition 1.2. Chain homotopic maps induce equal homomorphisms on homology.

Proof. For a cycle $z \in B_n$, the difference $g_n(z) - f_n(z) = d_{C,n+1}K_n(z)$ is a boundary. Consequently

$$H_n(f) = H_n(g) : H_n(B_{\bullet}) \rightarrow H_n(C_{\bullet}).$$

□

In particular, chain maps

$$f : B_{\bullet} \rightarrow C_{\bullet}, \quad g : C_{\bullet} \rightarrow B_{\bullet}$$

whose composites are chain homotopic to the identity maps induce inverse isomorphisms on homology.

1.2 The Prism Operator and Homotopy Invariance

Write $I = [0, 1]$. A homotopy of maps of spaces gives a chain homotopy by triangulating each singular prism.

Theorem 1.3. Suppose that

$$f, g : X \rightarrow Y, \quad h : X \times I \rightarrow Y$$

satisfy $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$. There are homomorphisms

$$P_n : C_n(X) \rightarrow C_{n+1}(Y) \quad (n \geq 0)$$

with $P_{-1} = 0$ and

$$\partial_{n+1}P_n + P_{n-1}\partial_n = g_{\#,n} - f_{\#,n} : C_n(X) \rightarrow C_n(Y). \quad (1)$$

Thus singular homology is a homotopy invariant functor in each degree:

$$H_n : \text{Top} \rightarrow \text{Ab}.$$

Proof. The prism operator is constructed in Definition 1.5; Proposition 2.4 proves (1). For a cycle z , that identity gives

$$g_{\#,n}(z) - f_{\#,n}(z) = \partial_{n+1}P_n(z).$$

The two images therefore represent the same homology class.

□

1.3 Cutting the Prism Into Simplices

Fix the base dimension $n \geq 0$. For each prism index $j = 0, \dots, n$, the j th prism simplex has the ordered vertices

$$(e_0, 0), \dots, (e_j, 0), (e_j, 1), \dots, (e_n, 1).$$

Write $s_i = (e_i, 0)$ for the bottom vertices. The affine parametrization is

$$\tau_j : |\Delta^{n+1}| \rightarrow |\Delta^n| \times I, \quad \tau_j(e_k) = \begin{cases} (e_k, 0), & 0 \leq k \leq j, \\ (e_{k-1}, 1), & j+1 \leq k \leq n+1. \end{cases}$$

Here k is a vertex position in the prism simplex. The e_k on the left is a standard vertex of the source; those on the right are standard vertices of the base. Affine extension gives the coordinate formula

$$\tau_j(t_0, \dots, t_{n+1}) = \left(\sum_{k=0}^j t_k e_k + \sum_{k=j+1}^{n+1} t_k e_{k-1}, \sum_{k=j+1}^{n+1} t_k \right).$$

Thus τ_j identifies the two source coordinates in positions $j, j+1$ in the base, and adds the coordinates after position j to obtain the height.

Remark 1.4. For a point (x, t) in the prism, now write $x = \sum_{i=0}^n t_i e_i$. It belongs to the j th piece exactly when

$$\sum_{i=j+1}^n t_i \leq t \leq \sum_{i=j}^n t_i. \quad (2)$$

Indeed, the unique coordinates in the source of τ_j are

$$\left(t_0, \dots, t_{j-1}, \sum_{i=j}^n t_i - t, t - \sum_{i=j+1}^n t_i, t_{j+1}, \dots, t_n \right).$$

They sum to 1, and they are nonnegative exactly under (2). An empty sum is zero, and an empty string of coordinates is omitted.

For a fixed base point, the endpoints of the intervals in (2) occur in order:

$$0 \leq t_n \leq t_{n-1} + t_n \leq \dots \leq t_0 + \dots + t_n = 1.$$

Hence the pieces cover the prism. For $j < j'$, membership in both pieces forces

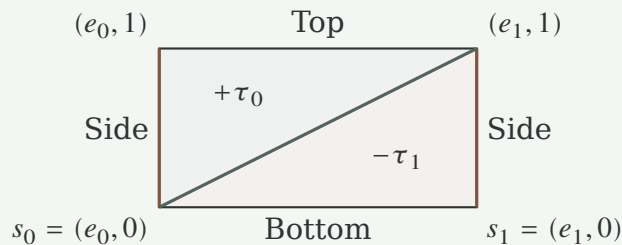
$$t = \sum_{i=j+1}^n t_i = \sum_{i=j'}^n t_i, \quad t_{j+1} = \dots = t_{j'-1} = 0.$$

Their intersection is exactly the common face whose ordered vertices are

$$(e_0, 0), \dots, (e_j, 0), (e_{j'}, 1), \dots, (e_n, 1).$$

Here a face means all linear combinations of the listed vertices with nonnegative coefficients summing to 1. Thus the pieces meet along faces and form a triangulation.

For $n = 1$ the picture is a square divided into two triangles:



The open diagonal is inside the square. The vertical edges lie over the boundary points of the base interval. This distinction persists in every dimension.

Definition 1.5. For a singular simplex σ , define σ_j by the composite

$$|\Delta^{n+1}| \xrightarrow{\tau_j} |\Delta^n| \times I \xrightarrow{\sigma \times \text{id}_I} X \times I \xrightarrow{h} Y.$$

Explicitly, the middle map is

$$\sigma \times \text{id}_I : |\Delta^n| \times I \rightarrow X \times I, \quad (x, t) \mapsto (\sigma(x), t).$$

The free abelian universal property extends the following assignment uniquely to a homomorphism:

$$P_n : C_n(X) \rightarrow C_{n+1}(Y), \quad \sigma \mapsto \sum_{j=0}^n (-1)^j \sigma_j.$$

2 The Boundary Identity and Absolute Homology

The boundary of the signed prism consists of its two ends and the prisms on its base faces. This identity gives homotopy invariance and the first absolute computations.

2.1 Prism Faces and Their Cancellation

The terms in the boundary are indexed by a Cartesian product:

$$(j, k) \in \{0, \dots, n\} \times \{0, \dots, n+1\}, \quad \partial_{n+1} P_n(\sigma) = \sum_{j=0}^n \sum_{k=0}^{n+1} (-1)^{j+k} \sigma_j d^k.$$

The first index chooses the prism piece; the second deletes its vertex in position k . Each corresponding affine face has type

$$\tau_j d^k : |\Delta^n| \rightarrow |\Delta^n| \times I.$$

The base vertex e_j appears twice in the piece, in positions j and $j+1$. Every other base vertex appears once.

Remark 2.1. For $\epsilon \in \{0, 1\}$ the endpoint inclusion of the base is

$$\iota_\epsilon : |\Delta^n| \rightarrow |\Delta^n| \times I, \quad x \mapsto (x, \epsilon).$$

The equalities of affine maps, including their signs, are

$$\begin{aligned} \tau_0 d^0 &= \iota_1 : |\Delta^n| \rightarrow |\Delta^n| \times I, & (-1)^{0+0} &= +1, \\ \tau_n d^{n+1} &= \iota_0 : |\Delta^n| \rightarrow |\Delta^n| \times I, & (-1)^{n+n+1} &= -1. \end{aligned}$$

After composition with the homotopy, these give $g\sigma - f\sigma$.

Remark 2.2. For $0 \leq j < n$, the two faces

$$\tau_j d^{j+1} = \tau_{j+1} d^{j+1} : |\Delta^n| \rightarrow |\Delta^n| \times I$$

have the same ordered vertices:

$$(e_0, 0), \dots, (e_j, 0), (e_{j+1}, 1), \dots, (e_n, 1).$$

They are equal as parametrized simplices, not merely as subsets. The equality is the commutative square

$$\begin{array}{ccc} |\Delta^n| & \xrightarrow{d^{j+1}} & |\Delta^{n+1}| \\ d^{j+1} \downarrow & & \downarrow \tau_j \\ |\Delta^{n+1}| & \xrightarrow{\tau_{j+1}} & |\Delta^n| \times I. \end{array}$$

Its relative interior lies inside the prism: all base coordinates are positive and both heights occur. Its boundary may meet the outer boundary.

The coefficients of this same ordered face in the two boundaries are

$$(-1)^j(-1)^{j+1} = -1, \quad (-1)^{j+1}(-1)^{j+1} = +1.$$

Thus the signed prism simplices induce opposite orientations on the face. Their contributions cancel, including after composition with h .

Remark 2.3. Deleting the unique vertex over e_i , with $i \neq j$, gives a side face over $t_i = 0$. Its position is i for $i < j$ and $i + 1$ for $i > j$. In the lower-dimensional prism the symbols τ_{j-1} and τ_j use the same affine definition, now with source $|\Delta^n|$ and target $|\Delta^{n-1}| \times I$. The two cases give the squares

$$\begin{array}{ccc} |\Delta^n| & \xrightarrow{d^i} & |\Delta^{n+1}| \\ \tau_{j-1} \downarrow & & \downarrow \tau_j \quad (i < j), \\ |\Delta^{n-1}| \times I & \xrightarrow{d^i \times \text{id}_I} & |\Delta^n| \times I \end{array}$$

$$\begin{array}{ccc} |\Delta^n| & \xrightarrow{d^{i+1}} & |\Delta^{n+1}| \\ \tau_j \downarrow & & \downarrow \tau_j \quad (i > j), \\ |\Delta^{n-1}| \times I & \xrightarrow{d^i \times \text{id}_I} & |\Delta^n| \times I \end{array}$$

Both squares commute by equality of ordered vertices. Omitting e_i changes the repeated vertex index from j to $j - 1$ when $i < j$ and leaves it unchanged when $i > j$.

Proposition 2.4. For the prism operator of Definition 1.5 and every $n \geq 0$, with $P_{-1} = 0$, one has

$$\partial_{n+1}P_n + P_{n-1}\partial_n = g_{\#,n} - f_{\#,n} : C_n(X) \rightarrow C_n(Y).$$

Proof. Fix a singular n -simplex σ with $n \geq 1$. Expand the two compositions:

$$\partial_{n+1}P_n(\sigma) = \sum_{j=0}^n \sum_{k=0}^{n+1} (-1)^{j+k} \sigma_j d^k \quad \text{Definition 1.5 and the boundary formula.}$$

$$P_{n-1}\partial_n(\sigma) = \sum_{i=0}^n \sum_{j=0}^{n-1} (-1)^{i+j} (\sigma d^i)_j \quad i \text{ selects the base face; } j \text{ its prism piece.}$$

Here $(\sigma d^i)_j$ denotes the j th prism simplex of the singular face σd^i . The endpoint terms in the first expansion are

$$\begin{aligned} +\sigma_0 d^0 &= h(\sigma \times \text{id}_I) \iota_1 = g\sigma & (j,k) &= (0,0); \text{ Remark 2.1.} \\ -\sigma_n d^{n+1} &= -h(\sigma \times \text{id}_I) \iota_0 = -f\sigma & (j,k) &= (n,n+1); \text{ Remark 2.1.} \end{aligned}$$

The remaining vertices have height 1 and 0, respectively. These are the two full base simplices; the second expansion consists of prisms on base faces.

The table matches every other term. The ordered pairs refer to the prism-piece index and deleted position in ∂P .

Indices in ∂P	Signed term	Opposite signed term
$(j, j+1), 0 \leq j < n$	$-\sigma_j d^{j+1}$	$+\sigma_{j+1} d^{j+1}$ in ∂P
$(j+1, i), 0 \leq i \leq j < n$	$(-1)^{i+j+1} \sigma_{j+1} d^i$	$(-1)^{i+j} (\sigma d^i)_j$ in $P\partial$
$(j, i+1), 0 \leq j < i \leq n$	$(-1)^{i+j+1} \sigma_j d^{i+1}$	$(-1)^{i+j} (\sigma d^i)_j$ in $P\partial$

In the first row the paired index is $(j+1, j+1)$, and equality of the two faces is Remark 2.2. The last two rows use the commuting squares in Remark 2.3; their index ranges partition all (i, j) with $0 \leq i \leq n$ and $0 \leq j < n$. Thus the $(n+1)(n+2)$ terms of ∂P consist of 2 endpoints, $2n$ interior terms, and $n(n+1)$ side terms. The interior terms cancel in pairs; the side terms match all $n(n+1)$ terms of $P\partial$ with opposite signs. Explicitly,

$$\begin{aligned}
\partial_{n+1} P_n(\sigma) &= g\sigma - f\sigma && \text{endpoint indices } (0, 0), (n, n+1). \\
&+ \sum_{j=0}^{n-1} (-\sigma_j d^{j+1} + \sigma_{j+1} d^{j+1}) && \text{interior pairs, first row.} \\
&- \sum_{j=0}^{n-1} \sum_{i=0}^j (-1)^{i+j} (\sigma d^i)_j && \text{side indices } (j+1, i), \text{ second row.} \\
&- \sum_{j=0}^{n-1} \sum_{i=j+1}^n (-1)^{i+j} (\sigma d^i)_j && \text{side indices } (j, i+1), \text{ third row.} \\
&= g\sigma - f\sigma - P_{n-1} \partial_n(\sigma) && \text{cancel interior pairs and join the side ranges.}
\end{aligned}$$

For $n = 0$, the prism is one interval and $\partial_1 P_0(\sigma) = g\sigma - f\sigma$, with $P_{-1} = 0$. Adding $P_{n-1} \partial_n$ and extending from generators proves (1) in every degree. $\square$

By the general homotopy-invariance proposition in Lecture 5, homotopy equivalences induce homology isomorphisms.

2.2 Components, Points, and Disjoint Unions

Write $\pi_0^{\text{path}}(X)$ for the set of path components, and $\mathbb{Z}[E]$ for the free abelian group on a set E .

Proposition 2.5. There is a natural isomorphism

$$\eta_X : H_0(X) \rightarrow \mathbb{Z}[\pi_0^{\text{path}}(X)], \quad [x] \mapsto \text{the generator for the component of } x.$$

Proof. Let B be an abelian group. A homomorphism

$$\alpha : C_0(X) \rightarrow B$$

is an assignment of an element of B to each point. The boundary of a path is its endpoint minus its starting point. Thus $\alpha \partial_1 = 0$ exactly when the assignment is constant on path components. The cokernel property gives the unique factorization

$$\begin{array}{ccccc}
C_1(X) & \xrightarrow{\partial_1} & C_0(X) & \xrightarrow{q} & H_0(X) \\
& & & \searrow \alpha & \downarrow \bar{\alpha} \\
& & & & B.
\end{array}$$

This is the free abelian universal property for the set of path components. It proves the isomorphism, and its description on points proves naturality. $\square$

Proposition 2.6. For the one-point space $*$,

$$H_n(*) = \begin{cases} \mathbb{Z}, & n = 0, \\ 0, & n > 0. \end{cases}$$

Proof. There is exactly one singular simplex in every degree. Hence $C_n(*) = \mathbb{Z}$, and for $n \geq 1$ the differential is

$$\partial_n : \mathbb{Z} \rightarrow \mathbb{Z}, \quad \partial_n(1) = \sum_{i=0}^n (-1)^i = \begin{cases} 1, & n > 0 \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

The chain complex is

$$\cdots \rightarrow \underset{3}{\mathbb{Z}} \xrightarrow{0} \underset{2}{\mathbb{Z}} \xrightarrow{1} \underset{1}{\mathbb{Z}} \xrightarrow{0} \underset{0}{\mathbb{Z}} \rightarrow 0,$$

where the labels below the groups are their degrees. Kernels modulo images give the asserted groups. $\square$

Proposition 2.7. For a family $(X_i)_{i \in J}$, the summand inclusions induce isomorphisms

$$\bigoplus_{i \in J} C_*(X_i) \xrightarrow{\cong} C_*\left(\coprod_{i \in J} X_i\right), \quad \bigoplus_{i \in J} H_n(X_i) \xrightarrow{\cong} H_n\left(\coprod_{i \in J} X_i\right).$$

Proof. Each singular simplex has connected nonempty domain, so its image lies in one summand of the disjoint union. The generators therefore split by summand, and every face stays in the same summand. This proves the chain isomorphism. In a direct sum every element has finite support, so both cycles and boundaries are direct sums of the cycles and boundaries in the summands. Taking their quotient proves the homology isomorphism. $\square$

The same argument splits singular chains over path components, even when those components are not open: a simplex has path connected domain and hence lies in one path component, with its subspace topology. In particular, for a discrete space E ,

$$H_0(E) = \mathbb{Z}[E], \quad H_n(E) = 0 \quad (n > 0).$$

For the empty space all chain and homology groups are zero.

2.3 Contractible Spaces and Deformation Retractions

A space is *contractible* if it is nonempty and its identity is homotopic to a constant map.

Proposition 2.8. For a contractible space X ,

$$H_0(X) \cong \mathbb{Z}, \quad H_n(X) = 0 \quad (n > 0).$$

Proof. Choose a point x_0 to which X contracts. The maps

$$\text{pr}_X : X \rightarrow *, \quad g : * \rightarrow X, \quad * \mapsto x_0$$

satisfy $\text{pr}_X g = \text{id}_*$ and $g \text{pr}_X \simeq \text{id}_X$. Their induced maps

$$(\text{pr}_X)_* : H_n(X) \rightarrow H_n(*), \quad g_* : H_n(*) \rightarrow H_n(X)$$

are therefore inverse isomorphisms. Apply the computation for a point. $\square$

For a nonempty convex subset $V \subseteq \mathbb{R}^m$ and $x_0 \in V$, the contraction is

$$h : V \times I \rightarrow V, \quad (x, t) \mapsto (1 - t)x + tx_0.$$

This applies to Euclidean spaces, balls, intervals, and standard simplices. It also applies when V is star-shaped about x_0 , meaning that it contains the line segment joining x_0 to each of its points.

A *deformation retraction* consists of an inclusion and a retraction

$$\iota : X' \hookrightarrow X, \quad r : X \rightarrow X'$$

and a homotopy $h : X \times I \rightarrow X$ satisfying

$$r\iota = \text{id}_{X'}, \quad h(x, 0) = x, \quad h(x, 1) = \iota r(x), \quad h(\iota(x'), t) = \iota(x').$$

Then ι and r are homotopy inverse maps, so

$$\iota_* : H_n(X') \rightarrow H_n(X), \quad r_* : H_n(X) \rightarrow H_n(X')$$

are inverse isomorphisms. For example,

$$r : \mathbb{R} \setminus \{0\} \rightarrow \{-1, 1\}, \quad x \mapsto x/|x|,$$

$$h : (\mathbb{R} \setminus \{0\}) \times I \rightarrow \mathbb{R} \setminus \{0\}, \quad (x, t) \mapsto (1 - t)x + tx/|x|$$

give a deformation retraction. The interpolation remains in the half-line containing x . Thus the homology is $\mathbb{Z}^2$ in degree zero and zero in positive degrees. Removing m distinct points from $\mathbb{R}$ similarly leaves $m + 1$ contractible components.

3 Relative Chains and the Interval

Quotient chain complexes give the relative exact sequence. Homotopies preserving the subspaces induce chain homotopies on these quotients.

3.1 Pairs, Quotients, and the Exact Sequence

A pair is a subspace inclusion

$$\iota : X' \hookrightarrow X.$$

Write (X, X') for this pair. The induced inclusion identifies the singular chains of X' with a subcomplex. Define

$$C_\bullet(X, X') = C_\bullet(X)/C_\bullet(X'), \quad H_n(X, X') = H_n(C_\bullet(X, X')).$$

A map of pairs is a commutative square

$$\begin{array}{ccc} X' & \xhookrightarrow{\iota_X} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xhookrightarrow{\iota_Y} & Y. \end{array}$$

The relative chain complex is a cokernel. For any chain complex $B_\bullet$ and chain map

$$\alpha : C_\bullet(X) \rightarrow B_\bullet, \quad \alpha \iota_\# = 0,$$

there is a unique induced map

$$\bar{\alpha} : C_\bullet(X, X') \rightarrow B_\bullet.$$

making the right square commute; the left square is a pushout:

$$\begin{array}{ccccc} C_*(X') & \xrightarrow{i_\#} & C_*(X) & \xrightarrow{\alpha} & B_* \\ \downarrow & & \downarrow q & & \parallel \\ 0 & \longrightarrow & C_*(X, X') & \xrightarrow{\bar{\alpha}} & B_* \end{array}$$

For a map of pairs this constructs the relative chain map in the square

$$\begin{array}{ccc} C_*(X) & \xrightarrow{q_X} & C_*(X, X') \\ f_\# \downarrow & & \downarrow f_\# \\ C_*(Y) & \xrightarrow{q_Y} & C_*(Y, Y'). \end{array}$$

Uniqueness gives compatibility with identities and composition, hence functoriality.

A relative cycle is represented by $c \in C_n(X)$ with $\partial_n c \in C_{n-1}(X')$. Two such representatives have the same relative homology class exactly when their difference is a boundary in X plus a chain in X' .

Theorem 3.1. Every pair has a natural exact sequence

$$\dots \rightarrow H_n(X') \xrightarrow{i_*} H_n(X) \xrightarrow{q_*} H_n(X, X') \xrightarrow{\delta_n} H_{n-1}(X') \rightarrow \dots$$

ending in

$$H_0(X') \xrightarrow{i_*} H_0(X) \xrightarrow{q_*} H_0(X, X') \rightarrow 0.$$

The connecting homomorphism is

$$\delta_n : H_n(X, X') \rightarrow H_{n-1}(X'), \quad [c] \mapsto [\partial_n c].$$

Proof. Apply the long exact sequence of chain complexes from Lecture 2 to

$$0 \rightarrow C_*(X') \xrightarrow{i_\#} C_*(X) \xrightarrow{q} C_*(X, X') \rightarrow 0.$$

For a relative cycle, $\partial_n c$ lies in the subcomplex and is a cycle because $\partial_{n-1} \partial_n = 0$. Replacing c by $c + \partial_{n+1} b + c'$ with $c' \in C_n(X')$ changes its boundary by $\partial_n c'$, so the connecting class is well defined. The chain-complex theorem supplies exactness and naturality. $\square$

Example 3.2. Take $X = I$ and $X' = \partial I = \{0, 1\}$. Contractibility of I and the computation for discrete spaces reduce the exact sequence to

$$0 \rightarrow H_1(I, \partial I) \xrightarrow{\delta_1} \mathbb{Z}^2 \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow H_0(I, \partial I) \rightarrow 0,$$

where

$$\varepsilon : \mathbb{Z}^2 \rightarrow \mathbb{Z}, \quad (a, b) \mapsto a + b.$$

Both endpoint generators map to the component generator of I . The map is surjective, and its kernel is $\mathbb{Z}(-1, 1)$. Thus

$$H_n(I, \partial I) = \begin{cases} \mathbb{Z}, & n = 1, \\ 0, & n \neq 1. \end{cases}$$

The singular edge

$$\sigma : |\Delta^1| \rightarrow I, \quad (t_0, t_1) \mapsto t_1$$

has boundary equal to the point generator at 1 minus the point generator at 0. Thus $\delta_1[\sigma] = (-1, 1)$ and $[\sigma]$ is a generator.

Reflection is the map of pairs induced by

$$\rho : I \rightarrow I, \quad t \mapsto 1 - t.$$

Its restriction to the endpoints is

$$\rho' : \partial I \rightarrow \partial I, \quad t \mapsto 1 - t.$$

Naturality gives

$$\begin{array}{ccc} H_1(I, \partial I) & \xrightarrow{\delta_1} & H_0(\partial I) \\ \rho_* \downarrow & & \downarrow \rho'_* \\ H_1(I, \partial I) & \xrightarrow{\delta_1} & H_0(\partial I). \end{array}$$

The right map exchanges the endpoint generators, sending $(-1, 1)$ to its negative. Injectivity of δ_1 therefore gives $\rho_*[\sigma] = -[\sigma]$.

3.2 Relative Homotopies and Reduced Homology

A homotopy preserving the subspaces descends to the quotient chain complexes.

Proposition 3.3. Suppose that

$$f, g : (X, X') \rightarrow (Y, Y')$$

are joined by a homotopy

$$h : X \times I \rightarrow Y, \quad h(X' \times I) \subseteq Y'.$$

Then the induced maps agree:

$$f_* = g_* : H_n(X, X') \rightarrow H_n(Y, Y').$$

Proof. For a simplex in X' , all its prism simplices have image in Y' . Hence the prism homomorphisms restrict and descend:

$$P_n : C_n(X') \rightarrow C_{n+1}(Y'), \quad \bar{P}_n : C_n(X, X') \rightarrow C_{n+1}(Y, Y').$$

Identity (1) descends to the quotient. It makes the relative chain maps chain homotopic, so they induce equal homology maps. $\square$

The subspace condition is essential. The contraction

$$h : I \times I \rightarrow I, \quad (x, t) \mapsto (1 - t)x$$

moves the endpoint 1 into the interior for $0 < t < 1$. It is therefore not a homotopy of the pair $(I, \partial I)$.

For a nonempty space X , the augmentation is the chain map

$$\varepsilon : C_\bullet(X) \rightarrow \mathbb{Z}[0],$$

whose degree-zero component is

$$\varepsilon_0 : C_0(X) \rightarrow \mathbb{Z}, \quad \sum_i a_i x_i \mapsto \sum_i a_i,$$

and whose positive-degree components are zero. Here $\mathbb{Z}[0]$ is concentrated in degree zero. Every path boundary has coefficient sum zero, so this is a chain map. It is surjective because X is nonempty.

Define reduced chains by the kernel, equivalently by the pullback:

$$\begin{array}{ccc} \tilde{C}_\bullet(X) & \longrightarrow & C_\bullet(X) \\ \downarrow & & \downarrow \varepsilon \\ 0 & \longrightarrow & \mathbb{Z}[0]. \end{array}$$

Their homology is $\tilde{H}_n(X)$. The kernel exact sequence gives

$$\tilde{H}_n(X) \cong H_n(X) \quad (n > 0), \quad \tilde{H}_0(X) = \ker(\varepsilon_* : H_0(X) \rightarrow \mathbb{Z}).$$

The prism homomorphisms restrict to the reduced complexes, so reduced homology is homotopy invariant.

If X is contractible and X' is nonempty, the exact sequence of the pair gives isomorphisms

$$\delta_n : H_n(X, X') \xrightarrow{\cong} \tilde{H}_{n-1}(X') \quad (n \geq 1), \quad H_0(X, X') = 0.$$

For $n \geq 2$ both adjacent positive-degree groups of X vanish. For $n = 1$ the image of δ_1 is the kernel of the augmentation on $H_0(X')$.

For example, let X' be a finite discrete subspace consisting of $m \geq 1$ points in a contractible space X . Then

$$H_1(X, X') \cong \ker(\varepsilon : \mathbb{Z}^m \rightarrow \mathbb{Z}) \cong \mathbb{Z}^{m-1},$$

where ε sums the coordinates; all other relative groups vanish. Choose one marked point and paths from it to the others. Their relative classes form a basis because their boundaries are a basis of the augmentation kernel. In particular, three marked points in an interval or a disk give relative first homology $\mathbb{Z}^2$.

For an open cover, the small-chain subdivision theorem identifies the homology of chains supported in individual cover members with singular homology. The resulting short exact sequence of chain complexes induces the Mayer-Vietoris sequence.